Class #9 - The Matrix Equation $Ax = b$

1. Properties of a linear transformation $T : \mathbf{R^m} \to \mathbf{R^n}$

   (a) $T(0) =$ __________. Why?
   Fill in the blank: If ________________________, then ________________________(can use as a test for __________).

   (b) $T(c_1u_1 + c_2u_2) =$ __________

   (c) $T(\sum_{i=1}^{m} c_iu_i) =$ __________, i.e.,
   T(linear combination of inputs) $=$ ________________________.
   This is called the *Superpositon principle.*

2. True or False: There is a nonzero matrix A and a nonzero vector x so that $Ax = 0$.

3. Let $A = \begin{bmatrix} -1 & 5 \\ 0 & 3 \\ 3 & 1.5 \end{bmatrix} \quad x = \begin{bmatrix} \sqrt{3} \\ e \end{bmatrix}$.
   Then $Ax \in$ __________.
   What is the 2nd entry of the product Ax? __________

   Note: The ith entry in the Matrix-vector product $=$ sum of the entrywise product between the ________________________of the matrix and __________.

4. Example: $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
   $Ax =$ __________. This matrix does not change the identity of the vector x.
   Note: To have $Ax = x$, what should be the size of A?

5. Def: The *zero matrix* is a matrix with __________entry equals to zero.
   The $n \times n$ *identity matrix* is the matrix whose diagonals are all __________and every other entries are __________.

6. Question: Let $A \in \mathbf{R^{m \times n}}$ and $x \in$ __________. What is a condition on A so that $Ax = b$ has a solution for *all* $b \in$ __________ (i.e., columns of A span __________)?

7. Theorem: Let $A \in \mathbf{R^{m \times n}}$ and $b \in \mathbf{R^m}$. The following are equivalent (TFAE):

   (a) $Ax = b$ is consistent for all b.

   (b) Every __________ of A has a pivot position.

   (c) Every vector in $\mathbf{R^m}$ is a _________________________ of the __________ of A.

   (d) Columns of A span __________.

8. Question: Given nonzero vectors $u_1, u_2, \cdots, u_p$ in $\mathbf{R^m}$, how do we check if $u_i \notin Span\{u_j : j = 1, \cdots, i-1\}$ for each $i = 2, \cdots, p$?

   Idea:
   Look at the matrix
   $$\begin{bmatrix} u_1 & u_2 \end{bmatrix}.$$
   Case 1: If $u_2 \in Span\{u_1\}$, then how many pivot positions? __________
   Case 2: If $u_2 \notin Span\{u_1\}$, then how many pivot positions? __________

   If Case 1 occurs, then $u_2 \in Span\{u_1\}$.
   If Case 2 occurs, then

   check the matrix
   $$\begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix}.$$
   If this matrix has __________ pivot positions, then $u_3 \notin Span\{u_1, u_2\}$.

Fill in the blank: $u_i \notin Span\{u_1, \cdots, u_{i-1}\}$ when the matrix __________has pivot positions in every __________.

9. Suppose $A \in \mathbf{R^{m\times n}}$. What is the maximum number of pivot positions that A can have? __________

So, span of the columns of A is a __________of __________, and the columns of A can never span $\mathbf{R^m}$ if m__________n.